Parallel Computers (Multiprocessors and Multicomputers)

CSE Sem-4

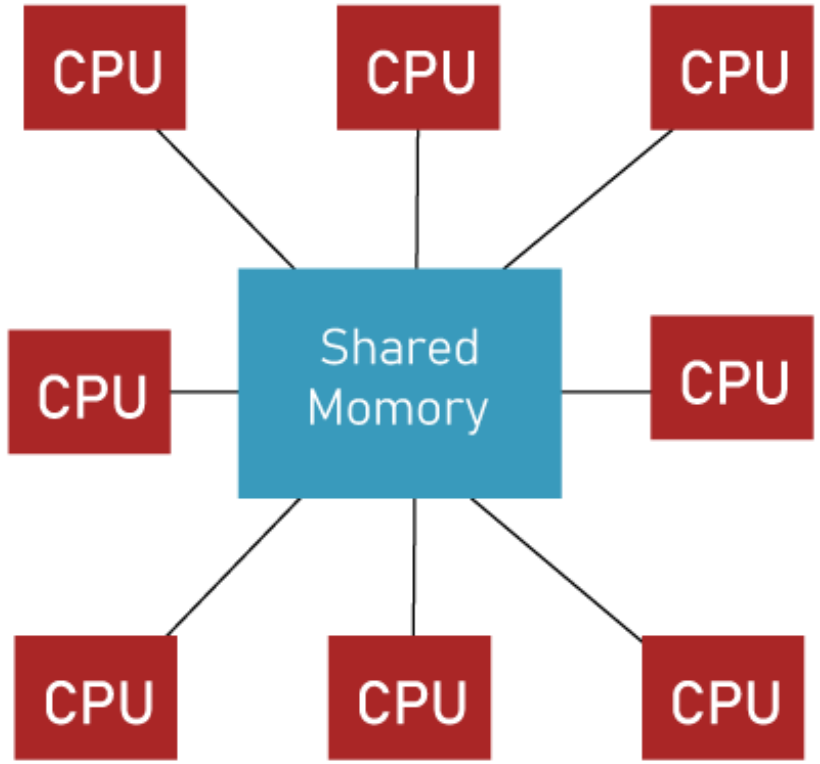
Introduction

- **A uniprocessor system,** also called a one-CPU system (**Von Neumann's architecture**): A system that utilizes a **single Central Processing Unit (CPU)** to execute instructions and perform tasks.
  - **Speed** can be **increased** by : (i) **Pipeline,** (ii) **Clock rate,** (iii) **Faster memory and Caches**
- **Parallel Computer:** A parallel computer is a computing system designed to **divide large, complex tasks into smaller pieces** and **process them simultaneously using multiple processors**.
  - **Two types: i) Multiple CPU** or **Multiprocessor** or **Shared memory multiprocessor**

    ii) Multicomputer or **Distributed memory multiprocessor**.
- A **multiprocessor** is a **single computer** that contains **numerous interconnected processors** with standard computing resources like memory and Input/output devices.
- The **multicomputer** is built by linking **numerous autonomous computers** via a **network**, and every autonomous system has its own computing resources, like memory and Input/output devices.

Shared-Memory Multiprocessors

- It is a system that **contains multiple processing units** (CPUs) that **share full access to a common Main Memory (RAM).**
- The processors **may communicate with each other** by reading and writing to the shared memory.
- Multiple processors may **execute various tasks simultaneously**. If one CPU has failed, it doesn't affect the tasks of the other processors. As a result, **a multiprocessor is more dependable**.
- Multiprocessors are called a **tightly coupled system**, due to a high degree of resource sharing.
- Various types of multiprocessors: **UMA** (Uniform Memory Access), **NUMA** (Non-Uniform Memory Access), **COMA** (Cache Only Memory Architecture).



Related Questions:

Q. Describe shared memory architecture. (4)

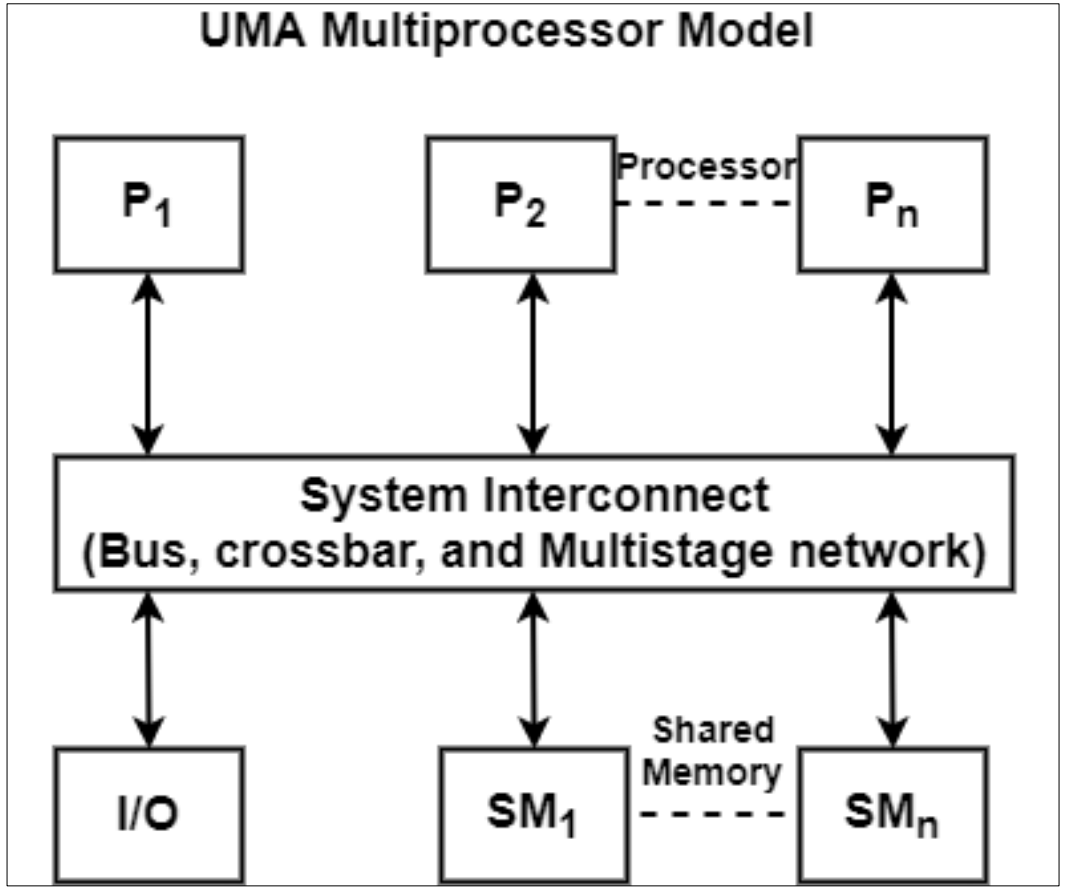
Q. Explain various types of shared memory multiprocessor architecture (5).

Q. Distinguish b/w loosely coupled and tightly coupled multiprocessor architectures. Which architecture is better and why? (5)

Models of Shared-Memory Multiprocessors

❖ ***UMA (Uniform Memory Access) Model:***

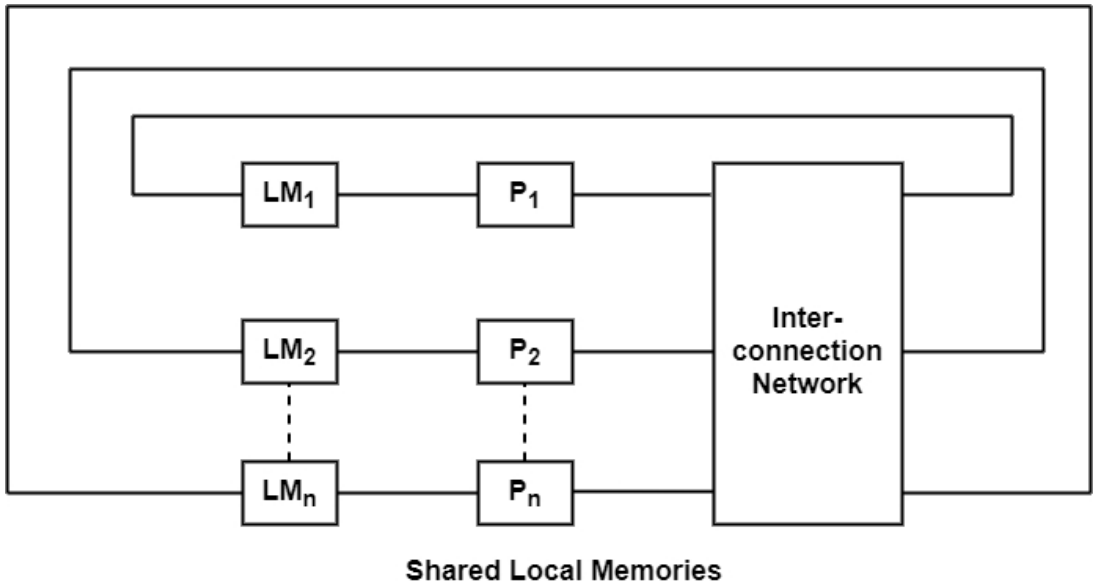
- In this model, all the **processors share a single physical memory uniformly**, and all **have an equal access time** to all the memory words.
- Each processor may **use a private cache**. Peripherals are also shared.
- UMA model is applicable for **time-sharing applications** by various users.
- To co-ordinate parallel execution, **synchronization and communication** among processors are done.
- As all the processors **have similar access to all peripheral devices**, the system is known as a **symmetric multiprocessor**.
- **E.g.** Sun Starfire Servers, Compaq alpha server, HP v series



Models of Shared-Memory Multiprocessors

❖ ***NUMA (Non-Uniform Memory Access) Model:***

- It is a **shared memory** system. The **shared memory is physically shared by all processors**, known as **local memories**.
- The collection of all local memories forms a **global address space** accessible by all processors.
- It is **faster to access a local memory with a local processor**.
- The **access of remote memory** attached to other processors **takes a longer time** because of the **added delay** through the interconnection network. So, **access time varies with the location** of the memory word.
- **E.g.** Cray, TC-2000, BBN and others



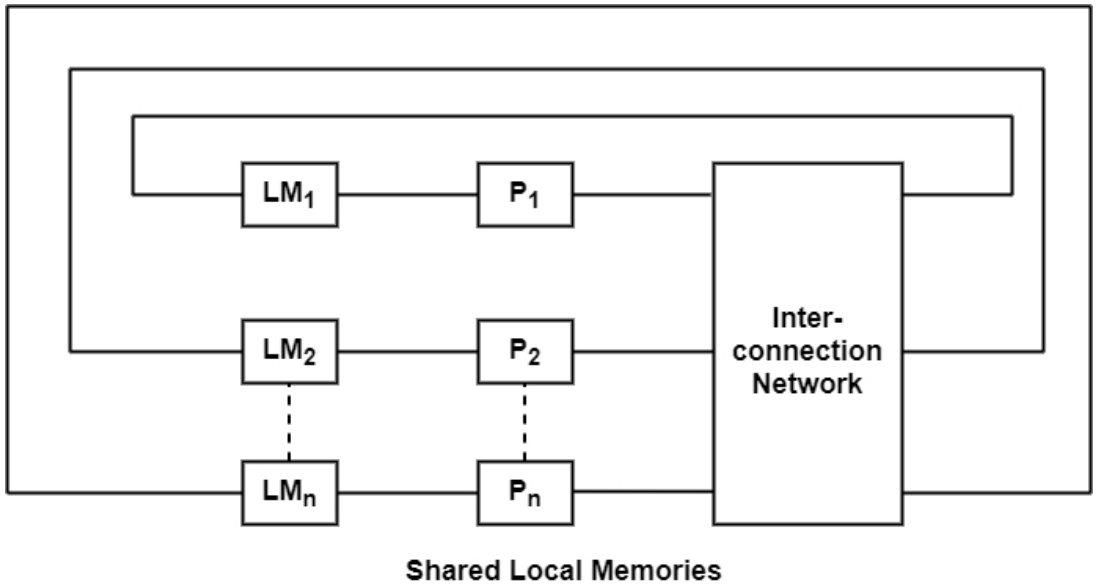
Related Questions:

Q. Compare and contrast between UMA and NUMA with example. (5)

Models of Shared-Memory Multiprocessors

❖ ***ccNUMA (Cache Coherent Non-Uniform Memory Access) Model:***

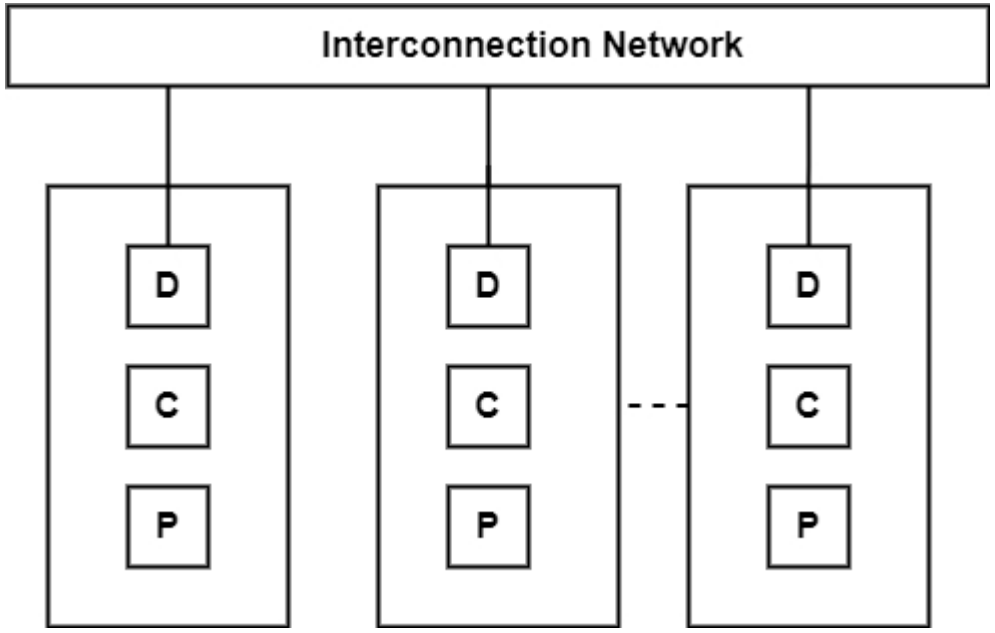
- Problem in NUMA: If multiple caches store **copies of the same memory location**, **inconsistencies** can occur when one processor updates its copy.
- **Cache coherence** is a property of multiprocessor systems that **ensures all caches have a consistent view of memory**.
- **Cache coherence** ensures that **all processors see the most up-to-date data.**
- Cache coherence is maintained by software, special-purpose hardware, or both.



Models of Shared-Memory Multiprocessors

❖ ***COMA (Cache Only Memory Architecture) Model:***

- This model is a special case of a NUMA machine**, where shared main memories are replaced with cache memory (C)**.
- There is **no memory hierarchy** at individual processor **(P)** node.
- All cache made a **global address space**.
- Remote cache access is assisted by the **distributed cache directories**. Cache directories **(D)** are the files where the data are stored temporarily.
- Depending on the interconnection network used, directories can be used to support in locating copies of cache blocks.

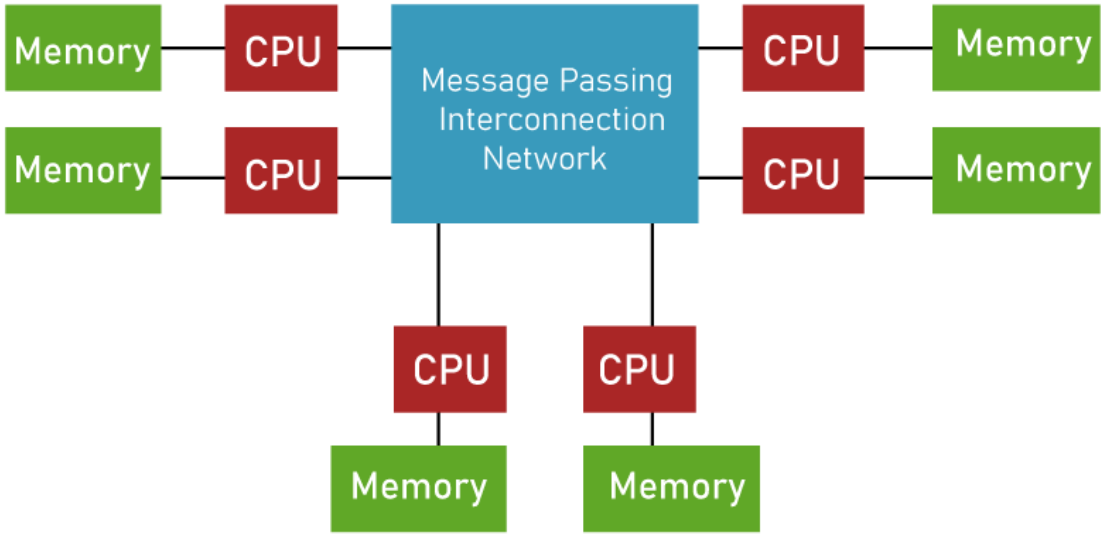


Q. What is memory dump?

A memory dump is taking all the information from RAM and creating a copy of it in our computer's hard drive. This process happens automatically when a computer crashes and right before the power turns off. Memory dumps help developers gather diagnostic information about the device crash, learn more about the event, and troubleshoot issues.

Distributed-Memory Multicomputers

- It is a **system consists of several individual nodes**.
- Each node is an **autonomous computer consisting of a processor, local memory and I/O devices**.
- An **interconnection message-passing network** allows the processors to communicate with one another.
- The message-passing network provides **point-to-point static connections** among the nodes.
- All **local memories are private** and are accessible only by local processors.
- For this, the multicomputers are called **"No-Remote Memory Access" (NORM)**.
- As it can transmit messages between the processors, the task may be categorized among the CPUs to be completed. Thus, the **multicomputer may be utilized for distributed computation**.
- As there **is no sharing of resources**, multicomputers are called **loosely coupled system.**



Q. What are the advantages and disadvantages of having a distributed memory? (5)

Q. Difference between centralized shared memory multiprocessor and distributed memory multiprocessor. (3)

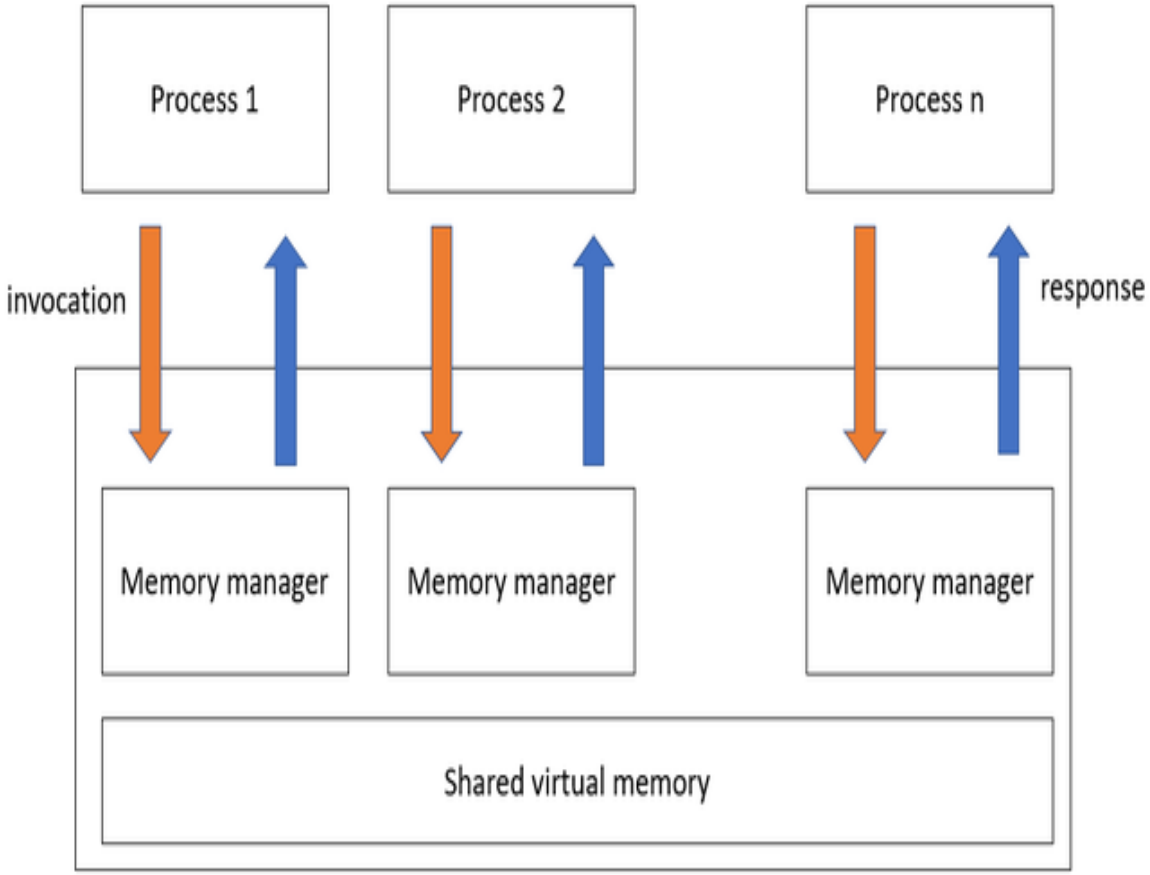
Q. Which architecture is better and why?

Loosely coupled system are most efficient when the interaction between tasks is minimal, whereas tightly coupled systems can tolerate a higher degree of interaction between tasks.

YEAR QA

Q. Distributed shared memory

DSM is a mechanism of allowing user processes to access shared data without using inter-process communications. In DSM, every node has its own memory and provides memory read and write services and it provides consistency protocols. The distributed shared memory (DSM) implements the shared memory model in distributed systems, but it doesn't have physical shared memory. All the nodes share the virtual address space provided by the shared memory model.



YEAR QA

Q. What are the different types of access in distributed shared memory architecture? (2)

On-Chip Memory:
- The data is present in the CPU portion of the chip.
- Memory is directly connected to address lines.
- It is expensive and complex.

Bus-Based Multiprocessors:
- A set of parallel wires called a bus acts as a connection between CPU and memory.
- Accessing of same memory simultaneously by multiple CPUs is prevented by using some algorithms
- Cache memory is used to reduce network traffic.

Ring-Based Multiprocessors:
- There is no global centralized memory
- All nodes are connected via a token passing ring.
- A single address line is divided into the shared area.

Difference between multiprocessors and multicomputers

Features	Multiprocessor System	Multicomputer System
Definition	It is a system **with multiple processors that enables programs to be processed at the same time.**	It is a collection of **processors linked by a communication network** that collaborate to solve a computation task.
Computing	It supports **parallel computing**.	It supports **distributed computing**.
Construction	It is **easy and less expensive** to develop.	It is **complex and expensive** to develop.
Type of network	It is a type of **dynamic network**.	It is a type of **static network**.
Communication between processing elements	It **requires proper communication** between the processing components and memory for successful resource allocation.	There is **no interaction** between processor units or memory resources.
Another name	It is also known as the **tightly coupled system**.	It is also known as **loosely coupled systems**.
Execution	It may execute the programs **very quickly**.	It may run **slowly**.